

Common-Target Content in a Primitive Möbius Tail: Row–Orbit Pruning, Cross-Factor Pruning, and the Coprime-Core Gate

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Abstract

Let

$$N_m(j) = mj + h, \quad N_n(j) = nj + h$$

be two positive affine targets in the primitive opened-row packet of the preceding Möbius-tail reduction. The orbit length is $J = X^{1-\beta+o(1)}$, the rows have size $Q = X^{\beta+o(1)}$, and $QJ \asymp X$; $\mathcal{D}$ below denotes the fixed packet data. Previous reflected-incidence bounds prune sectors with large common content in the selected divisor variables. They do not use the full target content

$$\mathcal{G}_{m,n}(j) := (N_m(j), N_n(j)).$$

On the primitive orbit we exploit the exact identity

$$\mathcal{G}_{m,n}(j) = (N_m(j), m - n),$$

so the full target gcd divides the row difference. For a fixed row pair, its exact values therefore range over only the divisors of $|m - n|$. Combining this divisor sparsity with the opened-row ℓ^1 mass gives, for each of the two mixed raw channels and the both-ultra raw channel, the unconditional bound

$$\mathcal{S}_{\mathcal{G}>Z} \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon \left(Q^2 + \frac{JQ^2}{Z} \right) \ll_{\varepsilon,h,\mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{Z} \right).$$

Thus $Z \geq X^\eta$ produces a fixed-power saving from the natural scale XQ , with exponent $\min\{\eta, 1 - \beta\}$ up to endpoint losses.

There is a second interpretation in the both-ultra channel. If

$$N_m = rU, \quad N_n = sV,$$

put $g = (r, s)$, $e = (U, V)$ and write $r = gr_0$, $s = gs_0$, $U = eU_0$, $V = eV_0$. Then, without any coprimality assumption between g and e ,

$$\mathcal{G}_{m,n}(j) = ge(r_0, V_0)(s_0, U_0) \mid m - n.$$

Consequently every sector with polynomially large cross-content on the left is power-small. This includes sectors with $(r, s) = 1$, which are outside the large reflected-content theorem.

At the high- β sample $\beta = 267/400$, take $r, s = X^{1/4+o(1)}$, $(r, s) = X^{o(1)}$, and $(U, V) = X^{2/5+o(1)}$. The reflected lcm is $X^{1/2+o(1)} > J$, while the preceding absolute-incidence method has zero saving on this negligible-reflected-content cell. The new target-content estimate has relaxed margin $133/400$: every fixed exponent below this value is available.

The argument is absolute and uses neither Möbius-sign cancellation nor a Chowla–Elliott input. It does not close the small-target-content core or the complete ultra-long residual, and it yields no prime-pair asymptotic. It isolates the next gate as signed dispersion on the small-target-content residual, with the fully coprime target sector as its sharp model.

Keywords: primitive Möbius tail; common target content; row-orbit incidence; cross-factor gcd; sparse fiber.

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1 Introduction

The ultra-long complement of a truncated divisor model is not made small merely by reflecting every long divisor across its target hyperbola. Reflection replaces a long divisor by a shorter complement and makes the orbit fibers visible, but the resulting family still contains enough short fibers to recover the natural scale. The preceding paper therefore used two kinds of shared content in the reflected variables: a large gcd on long fibers and a small primitive lcm on sparse fibers [5]. Those arguments are absolute and leave the small-reflected-content core.

This paper observes that the chosen divisor variables do not contain all of the available common-divisor information. For two physical rows $m \neq n$, put

$$N_m(j) = mj + h, \quad N_n(j) = nj + h, \quad \mathcal{G}_{m,n}(j) = (N_m(j), N_n(j)). \quad (1)$$

The full target gcd is independent of how either target is factored. On the primitive orbit it satisfies

$$\boxed{\mathcal{G}_{m,n}(j) = (N_m(j), m - n), \quad \mathcal{G}_{m,n}(j) \mid m - n.} \quad (2)$$

The divisibility statement appeared in the earlier target-gcd geometry for prime source rows [2]. The new point here is a weighted large-content pruning theorem for the actual opened rows $m = \ell d$, the orbit variable, and all three raw ultra channels.

Let Q be the row scale and J the orbit length, with

$$QJ \asymp X. \quad (3)$$

Let $\mathcal{I}$ be the physical orbit interval, of length $O_{\mathcal{D}}(J)$. For a fixed off-diagonal row pair, the exact values of $\mathcal{G}_{m,n}(j)$ belong to the divisor set of $|m - n|$. If $\mathcal{G}_{m,n}(j) = c$, the orbit variable lies in one residue class modulo c . It follows that

$$\#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m,n}(j) > Z\} \ll_{\varepsilon, \mathcal{D}} X^{\varepsilon} \left(1 + \frac{J}{Z}\right). \quad (4)$$

The physical row coefficients have one-row ℓ^1 mass $O(Q)$. Summing (4) over the row pairs gives the main estimate.

Theorem 1.1 (Common-target-content pruning, informal form). *For each of the two mixed raw channels and the both-ultra raw channel, the part on which $\mathcal{G}_{m,n}(j) > Z$ satisfies*

$$\boxed{|\mathcal{S}_{\mathcal{G} > Z}| \ll_{\varepsilon, h, \mathcal{D}} X^{\varepsilon} \left(Q^2 + \frac{JQ^2}{Z}\right).} \quad (5)$$

The same estimate holds for their sum.

Since $JQ^2 \asymp XQ$ is the natural absolute scale, the two terms on the right of (5) save respectively J^{-1} and Z^{-1} . Thus if

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad Z \geq X^{\eta}, \quad (6)$$

then every fixed

$$\eta_0 < \min\{\eta, 1 - \beta\} \quad (7)$$

is an available power saving. No cancellation between Möbius signs or between distinct rows is used.

The factorization consequence is especially useful. In the both-ultra channel write

$$N_m = rU, \quad N_n = sV. \quad (8)$$

Put

$$g = (r, s), \quad e = (U, V), \quad r = gr_0, \quad s = gs_0, \quad U = eU_0, \quad V = eV_0. \quad (9)$$

Then $(r_0, s_0) = (U_0, V_0) = 1$ and the stronger exact identity is

$$\boxed{\mathcal{G}_{m,n}(j) = ge(r_0, V_0)(s_0, U_0)}. \quad (10)$$

No assumption such as $(g, e) = 1$ is needed. The two final factors are the cross-contents left after the within-side gcfs are removed. Therefore (5) gives a power-saving upper bound for every retained sector on which the product in (10) exceeds Z . The factor $g = (r, s)$ is the reflected content used in TPC-29; $e = (U, V)$ and the two cross-contents are new to the present reduction's bookkeeping. In particular, the present estimate remains effective when $(r, s) = 1$.

At the exact high- β sample inherited from TPC-28,

$$\beta = \frac{267}{400}, \quad 1 - \beta = \frac{133}{400}, \quad (11)$$

consider the exponent cell

$$r, s = X^{1/4+o(1)}, \quad (r, s) = X^{o(1)}, \quad U, V = X^{3/4+o(1)}, \quad (U, V) = X^{2/5+o(1)}. \quad (12)$$

Its reflected lcm has exponent $1/2$, strictly above the orbit exponent $133/400$, and its primitive reflected lcm has the same exponent. Hence the TPC-29 absolute-incidence saving is zero on this cell. By contrast, the cross-content exponent is $2/5$, and the new relaxed margin is

$$\min \left\{ \frac{2}{5}, \frac{133}{400} \right\} = \boxed{\frac{133}{400}}. \quad (13)$$

This is an upper bound for a support-compatible formal sector, not an assertion that the sector carries nonzero arithmetic mass.

The result does not solve the residual problem. Once all polynomially large target content is removed, the remaining core has

$$\mathcal{G}_{m,n}(j) \leq X^\eta \quad (14)$$

for every fixed threshold used in the pruning, with the fully coprime case $\mathcal{G}_{m,n}(j) = 1$ as its sharp model. An absolute gcd argument has no further power saving there. The next plausible interface is a signed dispersion theorem across normalized determinant windows or determinant residue classes. Fixed-form logarithmically averaged two-point results such as [1] do not by themselves give a uniform estimate for the polynomially growing family retained here.

Section 2 records the physical packet and the three raw channels. Section 3 proves the exact gcd and orbit-class identities. Section 4 proves the weighted pruning theorem. Section 5 transfers it to cross-factor content, and section 6 gives the exact rational ledger. Section 7 separates the new region from earlier content and near-row reductions and formulates the next gate. Section 8 records the finite exact certificate and the claim boundary.

2 The physical packet and its raw channels

Fix $h \in \mathbb{Z} \setminus \{0\}$ and put $H = \text{rad } |h|$. We retain the primitive opened-row packet used in TPC-25–TPC-29 [3–5]. Its scales satisfy

$$QJ \asymp_{\mathcal{D}} X, \quad Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad 0 < \beta < 1. \quad (15)$$

Here $\mathcal{D}$ denotes the fixed smooth, residue, source, and support data.

The row labels are $\alpha = (\ell_\alpha, d_\alpha)$, where $\ell_\alpha \asymp L$ is prime, d_α is squarefree, and

$$m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathcal{D}} Q. \quad (16)$$

The size separation in the opened packet makes m_α determine α uniquely. In particular, the distinct row integers lie in an interval of length $O_{\mathcal{D}}(Q)$. The two row coefficients satisfy

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll_{\mathcal{D}} Q, \quad (17)$$

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L), \quad i = 1, 2. \quad (18)$$

Every contributing row and orbit point obeys

$$(m_\alpha, H) = 1, \quad (j, H) = 1. \quad (19)$$

For

$$N_\alpha(j) = m_\alpha j + h, \quad (20)$$

the joint support is contained in $O_{\mathcal{D}}(J)$ consecutive integers and

$$c_{\mathcal{D}} X \leq N_\alpha(j) \leq C_{\mathcal{D}} X. \quad (21)$$

Thus every target used below is positive. Moreover

$$(m_\alpha, h) = 1, \quad (j, h) = 1, \quad (22)$$

because H contains every prime divisor of h .

The row-pair mask is independent of all orbit and divisor variables and satisfies

$$|\mathbf{m}(\alpha, \gamma)| \leq 1, \quad \mathbf{m}(\alpha, \gamma) = 0 \quad \text{if } m_\alpha = m_\gamma. \quad (23)$$

The fixed-period factor $\Xi_{\alpha, \gamma}(j)$ and smooth weight $\mathcal{W}_{\alpha, \gamma}(j/J)$ are uniformly bounded. Their support may depend on the row pair, but it lies in an interval of $O_{\mathcal{D}}(J)$ consecutive integers.

2.1 The exact complement and pointwise envelopes

The inherited finite-model coefficient and the two residual coefficients are

$$\lambda'_R(w) := \begin{cases} w \sum_{b \leq R/w} \frac{\mu(wb)\mu(b)}{\varphi(wb)}, & w \leq R, \\ 0, & w > R, \end{cases} \quad (24)$$

$$a(w) = -\mu(w) \log w, \quad b_R(w) = a(w) - \lambda'_R(w). \quad (25)$$

They are supported on squarefree integers, and throughout the polynomial support used here,

$$|a(w)| + |b_R(w)| + |\lambda'_R(w)| \ll_{\mathcal{D}} (\log(2X))^3. \quad (26)$$

For $R \leq T < W_0$ and an integer $W_0 \asymp_{\mathcal{D}} X$ above every target in (21), define

$$A_{m,T}(j) := \sum_{\substack{w \leq T \\ w|mj+h}} b_R(w), \quad (27)$$

$$C_m(j) := \sum_{\substack{T < w \leq W_0 \\ w|mj+h}} a(w). \quad (28)$$

The elementary divisor bound and (26) give, uniformly on the physical support,

$$|A_{m,T}(j)| + |C_m(j)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon. \quad (29)$$

The exact calibrated cutoff identity in TPC-29 expands the joint difference into two mixed raw channels, one both-ultra raw channel, and two drift legs. The drift legs already carry the independent L^{-1} row saving [5]. The three raw kernels relevant here are

$$\begin{aligned} \mathcal{K}_{\alpha,\gamma}^{(1)}(j) &= A_{m_\alpha,T}(j)C_{m_\gamma}(j), \\ \mathcal{K}_{\alpha,\gamma}^{(2)}(j) &= C_{m_\alpha}(j)A_{m_\gamma,T}(j), \\ \mathcal{K}_{\alpha,\gamma}^{(3)}(j) &= C_{m_\alpha}(j)C_{m_\gamma}(j). \end{aligned} \quad (30)$$

For any condition $\mathcal{E}(\alpha, \gamma, j)$, define

$$\begin{aligned} \mathcal{S}_\nu(\mathcal{E}) &:= \sum_{\alpha,\gamma} \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathbf{m}(\alpha, \gamma) \sum_j \Xi_{\alpha,\gamma}(j) \mathcal{W}_{\alpha,\gamma}(j/J) \\ &\quad \times \mathcal{K}_{\alpha,\gamma}^{(\nu)}(j) \mathbf{1}_{\mathcal{E}(\alpha,\gamma,j)}, \quad 1 \leq \nu \leq 3. \end{aligned} \quad (31)$$

By (17), (29), and the orbit length, the natural absolute scale is

$$JQ^2 \asymp XQ. \quad (32)$$

The goal is to gain a fixed power from this scale on a condition defined by the full targets, before choosing or reflecting any of their divisors.

3 Full target content and orbit classes

The following identity is the arithmetic reason that target content can be summed through the row family. Its first divisibility consequence is the opened-row version of the target-gcd geometry in TPC-18 [2]; we include the short proof because the exact equality and the orbit class are used below.

Lemma 3.1 (Primitive target-gcd identity). *Let $m \neq n$ obey $(mn, h) = 1$, and let $(j, h) = 1$. Put*

$$N_m = mj + h, \quad N_n = nj + h, \quad \mathcal{G}_{m,n}(j) = (N_m, N_n). \quad (33)$$

Then

$$\boxed{\mathcal{G}_{m,n}(j) = (N_m, m - n) \mid |m - n|.} \quad (34)$$

In addition,

$$(\mathcal{G}_{m,n}(j), mn h) = 1. \quad (35)$$

Proof. Subtracting the two targets gives

$$(N_m, N_n) = (N_m, (n - m)j).$$

But

$$(N_m, j) = (mj + h, j) = (h, j) = 1.$$

The factor j may therefore be cancelled from the second argument, which proves (34). If a prime divides N_m and m , it divides h , contrary to $(m, h) = 1$; the same holds with n . Finally every prime dividing both N_m and h also divides mj and hence m or j after using the relevant primitivity. More directly, $(N_m, h) = (mj, h) = 1$ because $(m, h) = (j, h) = 1$. This proves (35). $\square$

Remark 3.2 (Why orbit primitivity is essential). Without $(j, h) = 1$, the equality in (34) need not hold: a factor shared by j and h can remain in both targets without dividing $m - n$. The nonprimitive content was treated separately by endpoint collapse in TPC-18. No such sector is silently included in the theorem below.

The exact target gcd selects a row congruence and a single orbit class.

Lemma 3.3 (Exact-content orbit occupancy). *Under the hypotheses of lemma 3.1, suppose*

$$\mathcal{G}_{m,n}(j) = c. \quad (36)$$

Then

$$m \equiv n \pmod{c}, \quad j \equiv -h\bar{m} \pmod{c}, \quad (37)$$

where $\bar{m}$ is the inverse of m modulo c . Consequently, in any interval $\mathcal{I}$ of at most $C_{\mathcal{D}}J$ consecutive integers,

$$\#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m,n}(j) = c\} \ll_{\mathcal{D}} 1 + \frac{J}{c}. \quad (38)$$

Proof. The first congruence follows from $c \mid m - n$ in (34). Since $c \mid N_m$ and $(c, m) = 1$ by (35), the divisibility $c \mid mj + h$ selects the second congruence. One residue class modulo c occupies an interval of the stated length at most $O_{\mathcal{D}}(1 + J/c)$ times. Requiring the gcd to equal c , rather than merely to be divisible by c , can only reduce this number. $\square$

Remark 3.4 (Exact layers prevent multiplicity loss). The proof of the main theorem groups each orbit point by the single integer $c = \mathcal{G}_{m,n}(j)$. It does not sum the condition $c \mid \mathcal{G}_{m,n}(j)$ over every divisor c , so no additional divisor multiplicity is hidden in the content decomposition.

4 Row-orbit energy and common-target pruning

For $c \geq 1$, define the absolute row congruence mass

$$\mathcal{R}_c := \sum_{\alpha, \gamma} |\gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)}| \mathbf{1}_{c \mid m_{\alpha} - m_{\gamma}}. \quad (39)$$

The following estimate was used for reflected content in TPC-29. We record it in the present notation because it is now paired with the full target gcd rather than a selected divisor gcd.

Lemma 4.1 (Opened-row residue energy). *Uniformly for $c \geq 1$,*

$$\boxed{\mathcal{R}_c \ll_{\mathcal{D}} \left(\frac{Q^2}{c} + Q \right) \log(2L)}. \quad (40)$$

Proof. For $i = 1, 2$ and $a \pmod{c}$, put

$$X_i(a) = \sum_{m_{\alpha} \equiv a \pmod{c}} |\gamma_{\alpha}^{(i)}|, \quad E_i(a) = \sum_{m_{\alpha} \equiv a \pmod{c}} |\gamma_{\alpha}^{(i)}|^2.$$

The row integers are distinct and lie in an interval of length $O_{\mathcal{D}}(Q)$. Each residue class contains $O_{\mathcal{D}}(Q/c + 1)$ rows, so Cauchy–Schwarz gives

$$X_i(a)^2 \ll_{\mathcal{D}} \left(\frac{Q}{c} + 1 \right) E_i(a).$$

It follows that

$$\begin{aligned} \mathcal{R}_c &= \sum_{a(c)} X_1(a) X_2(a) \\ &\ll_{\mathcal{D}} \left(\frac{Q}{c} + 1 \right) \left(\sum_a E_1(a) \right)^{1/2} \left(\sum_a E_2(a) \right)^{1/2}. \end{aligned}$$

Apply the two estimates in (18). □

For a row pair and orbit point, abbreviate

$$\mathcal{G}_{\alpha, \gamma}(j) := (N_{\alpha}(j), N_{\gamma}(j)). \quad (41)$$

Taking absolute values in any row channel gives

$$|\mathcal{K}_{\alpha, \gamma}^{(\nu)}(j)| \ll_{\varepsilon, \mathcal{D}} X^{\varepsilon}, \quad 1 \leq \nu \leq 3, \quad (42)$$

by (29).

Proposition 4.2 (A uniform exact-content layer). *For every integer $c \geq 1$ and $1 \leq \nu \leq 3$,*

$$\boxed{|\mathcal{S}_{\nu}(\mathcal{G}_{\alpha, \gamma}(j) = c)| \ll_{\varepsilon, h, \mathcal{D}} X^{\varepsilon} \left(1 + \frac{J}{c} \right) \left(\frac{Q^2}{c} + Q \right)}. \quad (43)$$

Proof. The mask in (23) first deletes the diagonal rows. By lemma 3.1, the exact layer is empty unless $c \mid m_{\alpha} - m_{\gamma}$. For each surviving row pair, lemma 3.3 gives $O_{\mathcal{D}}(1 + J/c)$ orbit points. Apply (42), sum the row weights through $\mathcal{R}_c$, and use (40). □

Summing (43) over every integer $c > Z$ would introduce an avoidable endpoint term. The stronger tail estimate comes from first fixing the row pair.

Lemma 4.3 (Fixed-row large-content occupancy). *Let $m \neq n$ be physical rows, let $\mathcal{I}$ be an interval of $O_{\mathcal{D}}(J)$ consecutive integers, and let $Z \geq 1$. Then*

$$\boxed{\#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m, n}(j) > Z\} \ll_{\varepsilon, h, \mathcal{D}} X^{\varepsilon} \left(1 + \frac{J}{Z} \right)}. \quad (44)$$

Proof. Put $\Delta = |m - n| \neq 0$. Every exact value $c = \mathcal{G}_{m, n}(j)$ divides Δ by lemma 3.1. Grouping by the exact value and applying lemma 3.3 gives

$$\begin{aligned} \#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m, n}(j) > Z\} &\leq \sum_{\substack{c \mid \Delta \\ c > Z}} O_{\mathcal{D}} \left(1 + \frac{J}{c} \right) \\ &\ll_{\mathcal{D}} \tau(\Delta) \left(1 + \frac{J}{Z} \right). \end{aligned} \quad (45)$$

Since $\Delta \ll_{\mathcal{D}} Q \ll X^{O_{\mathcal{D}}(1)}$, the standard divisor bound absorbs $\tau(\Delta)$ into X^{ε} . □

Theorem 4.4 (Common-target-content pruning). *Let $Z \geq 1$. For each $1 \leq \nu \leq 3$, the physical row channel in (31), restricted by*

$$\mathcal{E}_Z(\alpha, \gamma, j) := \{\mathcal{G}_{\alpha, \gamma}(j) > Z\}, \quad (46)$$

satisfies

$$|\mathcal{S}_\nu(\mathcal{E}_Z)| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^2 \left(1 + \frac{J}{Z}\right) \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{Z}\right). \quad (47)$$

The same bound holds for the sum of the three row channels.

Proof. The off-diagonal mask gives $m_\alpha \neq m_\gamma$, so lemma 4.3 applies to every row pair. Use (42) and then the row ℓ^1 bounds:

$$\begin{aligned} |\mathcal{S}_\nu(\mathcal{E}_Z)| &\ll X^\varepsilon \left(1 + \frac{J}{Z}\right) \sum_{\alpha, \gamma} |\gamma_\alpha^{(1)} \gamma_\gamma^{(2)}| \\ &\ll X^\varepsilon Q^2 \left(1 + \frac{J}{Z}\right). \end{aligned}$$

Finally use $JQ \asymp X$. If $Z \gg_{\mathcal{D}} Q$, the event is empty because

$$\mathcal{G}_{\alpha, \gamma}(j) \mid m_\alpha - m_\gamma \ll_{\mathcal{D}} Q. \quad (48)$$

□

Corollary 4.5 (Power-saving form). *Suppose (15) holds and $Z \geq X^\eta$ for a fixed $\eta > 0$. Then, for every fixed*

$$\eta_0 < \min\{\eta, 1 - \beta\}, \quad (49)$$

one has

$$\sum_{\nu=1}^3 |\mathcal{S}_\nu(\mathcal{E}_Z)| \ll_{\eta, \eta_0, h, \mathcal{D}} XQX^{-\eta_0}. \quad (50)$$

Proof. Relative to $XQ \asymp JQ^2$, the two terms in (47) have respective ratios J^{-1} and Z^{-1} . Use (15); the strict gap in (49) absorbs logarithms and the $o(1)$ scale factors. □

Remark 4.6 (The absolute one-point floor). The term Q^2 comes from the $+1$ in the occupancy of each exact content class. It is power-small because J is a fixed power, but the present argument does not replace it by JQ . This is a boundary of the absolute fixed-row proof, not an arithmetic lower bound or an optimality statement.

5 Four-factor content and cross-factor pruning

The full target gcd can be read from any pair of target factorizations, but it is not usually the product of the two within-side gcds alone.

Proposition 5.1 (Exact four-factor content decomposition). *Let*

$$N_m = rU, \quad N_n = sV \quad (51)$$

with positive integers r, s, U, V . Put

$$g = (r, s), \quad e = (U, V), \quad (52)$$

and write

$$r = gr_0, \quad s = gs_0, \quad U = eU_0, \quad V = eV_0. \quad (53)$$

Thus $(r_0, s_0) = (U_0, V_0) = 1$. Define the two normalized cross-contents

$$x = (r_0, V_0), \quad y = (s_0, U_0). \quad (54)$$

Then

$$(N_m, N_n) = gexy. \quad (55)$$

In particular, on the primitive physical orbit,

$$gexy = \mathcal{G}_{m,n}(j) \mid m - n. \quad (56)$$

No coprimality between g and e is asserted or required.

Proof. Factoring the two common within-side contents gives

$$(N_m, N_n) = ge(r_0 U_0, s_0 V_0). \quad (57)$$

We prove

$$(r_0 U_0, s_0 V_0) = (r_0, V_0)(s_0, U_0) \quad (58)$$

prime by prime. Fix a prime p and write

$$a = v_p(r_0), \quad b = v_p(s_0), \quad u = v_p(U_0), \quad v = v_p(V_0).$$

The two primitive conditions give

$$\min\{a, b\} = 0, \quad \min\{u, v\} = 0.$$

If $a > 0$, then $b = 0$. When $u > 0$, one has $v = 0$ and both sides of

$$\min\{a + u, b + v\} = \min\{a, v\} + \min\{b, u\} \quad (59)$$

are zero; when $v > 0$, one has $u = 0$ and both sides are $\min\{a, v\}$. If $u = v = 0$, both sides are again zero. The case $b > 0$ is symmetric, and the case $a = b = 0$ is immediate. This proves (58). Equations (57) and (34) now give the result. $\square$

Remark 5.2 (Why the cross terms cannot be dropped). Take $r = 2$, $U = 3$, $s = 3$, and $V = 4$. Then $g = e = 1$, but

$$(rU, sV) = (6, 12) = 6.$$

The missing factors are exactly $x = (2, 4) = 2$ and $y = (3, 3) = 3$. Thus the generally valid statement is not $(N_m, N_n) = ge$, but the four-factor identity (55).

5.1 Expanded divisor channels

Each kernel in (30) is a finite sum over one divisor of each target. In a mixed channel, one selected divisor comes from the prefix and the other from the ultra complement; in the both-ultra channel both are reflected complement variables. For any chosen pair $r \mid N_m$, $s \mid N_n$, put

$$U = N_m/r, \quad V = N_n/s. \quad (60)$$

Indeed, for a prefix leg and a reflected ultra leg, respectively,

$$\sum_{r \mid N_m} |b_R(r)| \ll \tau(N_m)(\log(2X))^3, \quad (61)$$

$$\sum_{r \mid N_m} |a(N_m/r)| \ll \tau(N_m) \log(2X). \quad (62)$$

The same estimates hold for N_n . Since both targets are $O_{\mathcal{D}}(X)$, the divisor bound shows that the absolute product mass of any mixed or both-ultra subfamily is bounded by the full pointwise divisor envelope. Here $\kappa^{(\nu)}(j, r, s)$ denotes the product of the two divisor coefficients in the expanded ν -th kernel, so

$$\sum_{\substack{r|N_m, s|N_n \\ \text{retained channel supports}}} |\kappa^{(\nu)}(j, r, s)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon. \quad (63)$$

Consequently the proof of [theorem 4.4](#) applies unchanged after restricting the expanded divisor sum to any condition implying $\mathcal{G}_{m,n}(j) > Z$.

Corollary 5.3 (Cross-factor pruning). *For each mixed or both-ultra expanded channel, restrict its individual divisor terms by any one of*

$$ge > Z, \quad e > Z, \quad x > Z, \quad y > Z, \quad \text{or more generally} \quad gexy > Z. \quad (64)$$

The resulting physical contribution satisfies

$$|\mathcal{S}_{\text{cross} > Z}| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon \left(Q^2 + \frac{XQ}{Z} \right). \quad (65)$$

If $Z \geq X^\eta$, every fixed $\eta_0 < \min\{\eta, 1 - \beta\}$ is a power-saving exponent.

Proof. By [proposition 5.1](#), each of the first four conditions in (64) implies $\mathcal{G}_{m,n}(j) > Z$, while the last is equivalent to that condition. Use (63) in place of (42) in the proof of [theorem 4.4](#). $\square$

5.2 What is new relative to reflected content

TPC-29 tracks

$$g = (r, s), \quad q = [r, s],$$

and proves a power-saving bound for a long-fiber sector with large g and a sparse sector in which the primitive reflected lcm q/g is small. The factors

$$e = (U, V), \quad x = (r/g, V/e), \quad y = (s/g, U/e)$$

are invisible if the two selected divisor variables are considered without their co-divisors. Thus [corollary 5.3](#) can be effective with $g = 1$ and q/g large, exactly where the preceding absolute reflected-incidence argument reaches only the natural scale.

No squarefreeness of r or s is used. In an ultra term the co-divisor coefficient $a(U)$ or $a(V)$ vanishes unless the relevant co-divisor is squarefree, but the four-factor identity itself is an integer identity and requires no coefficient support.

6 The high- β cross-content cell

We now insert the target-content theorem into the exact parameter sample closed through the selected cutoff in TPC-28. The inherited exponents are

$$\beta = \frac{267}{400}, \quad j = 1 - \beta = \frac{133}{400}, \quad t = \frac{193}{500}. \quad (66)$$

Thus

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (67)$$

and every reflected complement variable lies below

$$\frac{X}{T} = X^{307/500+o(1)}. \quad (68)$$

6.1 A reflected pair with negligible common content

In the both-ultra channel choose

$$r, s = X^{1/4+o(1)}, \quad g = (r, s) = X^{o(1)}. \quad (69)$$

The reflected lcm and its primitive part have exponents

$$[r, s] = X^{1/2+o(1)}, \quad (70)$$

$$\frac{[r, s]}{(r, s)} = X^{1/2+o(1)}. \quad (71)$$

Hence the fiber is sparse by

$$\frac{1}{2} - \frac{133}{400} = \boxed{\frac{67}{400}}. \quad (72)$$

However, the TPC-29 absolute-incidence saving on this cell is

$$\left(\frac{133}{400} - \frac{1}{2}\right)_+ = \boxed{0}. \quad (73)$$

Its content-rich sparse condition also fails because the primitive lcm in (71) is larger than J by a fixed power. The long-fiber theorem does not apply because $[r, s] > J$.

Since each target is $X^{1+o(1)}$, the two co-divisors have size

$$U, V = X^{3/4+o(1)} > T. \quad (74)$$

Impose the quotient-content condition

$$e = (U, V) = X^{2/5+o(1)}. \quad (75)$$

Then

$$U/e, V/e = X^{7/20+o(1)}. \quad (76)$$

To give the exponent cell a literal meaning, for fixed $\delta > 0$ let $\mathcal{B}_\delta(X)$ be the expanded both-ultra subfamily on which

$$X^{1/4-\delta} < r, s \leq X^{1/4+\delta}, \quad g \leq X^\delta, \quad (77)$$

$$X^{3/4-\delta} < U, V \leq X^{3/4+\delta}, \quad X^{2/5-\delta} < e \leq X^{2/5+\delta}. \quad (78)$$

A sequence of dyadic blocks has the exponents displayed in (69), (74), and (75) precisely when it belongs to $\mathcal{B}_\delta(X)$ eventually for every fixed $\delta > 0$. By [proposition 5.1](#), every term in $\mathcal{B}_\delta(X)$ satisfies $\mathcal{G}_{m,n}(j) \geq e$ and hence lies in the event

$$\mathcal{G}_{m,n}(j) > X^{2/5-\delta}. \quad (79)$$

Corollary 6.1 (High- β cross-content relaxed margin). *For every fixed $\eta_0 < 133/400$, there is a $\delta_0 > 0$ such that, for every fixed $0 < \delta < \delta_0$, the expanded physical both-ultra subsum $\mathcal{S}_{\text{high,cross}}(\delta)$ over $\mathcal{B}_\delta(X)$ satisfies*

$$|\mathcal{S}_{\text{high,cross}}(\delta)| \ll_{\delta, \eta_0, h, \mathcal{D}} X Q X^{-\eta_0}. \quad (80)$$

Equivalently, as $\delta \rightarrow 0$, the supremal relaxed exponent in this ledger is $133/400$; the endpoint itself is not asserted.

Proof. Choose δ_0 small enough that

$$\delta_0 < \min \left\{ \frac{2}{5} - \eta_0, \frac{67}{1600} \right\}. \quad (81)$$

The second restriction also gives $q/g = rs/g^2 \geq X^{1/2-4\delta} > J$ and hence $q > J$ for all sufficiently large X , so the strict separation from the TPC-29 regions is uniform on this window. On $\mathcal{B}_\delta(X)$ use [corollary 5.3](#) with $Z = X^{2/5-\delta}$. The two terms in (47), relative to XQ , save

$$1 - \beta = \frac{133}{400}, \quad \frac{2}{5} - \delta. \quad (82)$$

The first is $133/400 + o(1)$ at the inherited physical scale, and both exponents exceed η_0 for all sufficiently large X . This proves the claimed bound. Letting $\delta \rightarrow 0$ gives the stated relaxed margin. $\square$

6.2 Consistency checks

The block is a support-compatible formal exponent cell. First, $1/4 < 307/500$, so both reflected variables obey (68); and $3/4 > 193/500$, so both co-divisors are genuinely beyond the selected cutoff. Second, the necessary divisibility $e \mid m - n$ is compatible with the row scale because

$$\frac{267}{400} - \frac{2}{5} = \boxed{\frac{107}{400}} > 0. \quad (83)$$

On the optional macroscopically separated row subcell $|m - n| = X^{\beta+o(1)}$, the reduced target difference $(N_m - N_n)/e = ((m - n)j)/e$ and each reduced product have exponent

$$\left(\frac{267}{400} - \frac{2}{5} \right) + \frac{133}{400} = \frac{3}{5}, \quad (84)$$

$$\frac{1}{4} + \frac{7}{20} = \frac{3}{5}. \quad (85)$$

Thus, on that subcell, the row-orbit and factorization ledgers agree.

Nor is this cell a relabeling of the TPC-29 reflected-content cell. Moving the quotient gcd into both reflected variables would produce the exponent

$$\frac{1}{4} + \frac{2}{5} = \frac{13}{20} > \frac{307}{500}, \quad (86)$$

above the reflected ceiling, while the reduced co-divisor exponent $7/20$ lies below $t = 193/500$. The quotient content therefore crosses the selected cutoff and cannot be absorbed into the old reflected variables while preserving their channel roles.

Table 1: Exact ledger for the high- β cross-content cell.

Quantity	Exact value	Role
Orbit exponent j	$133/400$	natural fiber length
Reflected exponent	$1/4$	each of r, s
Reflected lcm exponent	$1/2$	sparse by $67/400$
Old incidence saving	0	primitive reflected core
Co-divisor exponent	$3/4$	both-ultra support
Quotient-content exponent	$2/5$	new full-target input
Reduced co-divisor exponent	$7/20$	after removing e
Row/content compatibility gap	$107/400$	e can divide $m - n$
Relaxed margin	$133/400$	every smaller fixed exponent is available

The corollary is an upper bound for this support-compatible formal block. It does not assert that the block is nonempty for every X , or that it carries a positive proportion of the original residual.

7 Comparison, the coprime core, and the next gate

7.1 The inherited identity and the new estimate

TPC-18 already observed that, on a primitive common orbit, the gcd of two affine targets divides the source-row difference [2]. The Euclidean step behind (34) is therefore inherited. The present additions are:

- (i) the exact identity for the actual composite opened rows $m = \ell d$ and their physical primitive support;
- (ii) the fixed-row large-content occupancy (44);
- (iii) the weighted family estimate (47) for all three raw channels;
- (iv) the exact four-factor identity (55) and its application to quotient-rich and cross-rich divisor sectors.

Thus there is no claim that the Euclidean gcd observation was unknown. The contribution within the present reduction is the physical family-level pruning bound obtained by placing that observation before divisor-variable separation.

7.2 Strict separation from the TPC-29 regions

Write

$$q = [r, s], \quad g = (r, s).$$

TPC-29 proves power saving in the two regions

$$q \leq J, \quad g > X^\eta, \tag{87}$$

$$q \geq J, \quad q/g \leq JX^{-\eta}. \tag{88}$$

The new theorem depends instead on the full target content

$$\mathcal{G} = gexy.$$

It is independent of whether q is below or above J . In particular, the high- β block in section 6 has $q > J$, $g = X^{o(1)}$, and $q/g > J$, so neither (87) nor (88) applies. Its large quotient content e nevertheless makes it power-small here.

The result is also distinct from deleting near-diagonal rows. The condition $\mathcal{G} > Z$ implies that $m - n$ has a large divisor; it does not imply that $|m - n|$ is small. The high- β cell is compatible with $|m - n| = X^{\beta+o(1)}$, far from the near-row range.

7.3 The small-content residual

After applying theorem 4.4 with a fixed threshold X^η , what remains satisfies

$$\boxed{\mathcal{G}_{m,n}(j) \leq X^\eta}. \tag{89}$$

The sharp model is the fully coprime target sector

$$(N_m(j), N_n(j)) = 1. \tag{90}$$

There

$$g = e = x = y = 1$$

for every selected factorization. Neither shared content nor a repartition of the same factors supplies a power saving.

For a reflected pair, compatibility still gives

$$g \mid m - n.$$

The normalized row determinant

$$\Delta_0 = \frac{m - n}{g} \tag{91}$$

is therefore an integer. The next useful decomposition is by the number of allowed values of Δ_0 in translated windows or residue classes, not by another neighborhood of $\Delta_0 = 0$. Near-row windows have already been removed in the earlier pipeline; the genuinely new question concerns macroscopically separated rows whose normalized determinants occupy a thin additive spectrum.

An absolute estimate for one thin determinant class would not by itself close (89): summing all residue classes absolutely restores the natural scale. For the present strategy, the next proposed gate is a signed family estimate that recombines these classes while retaining part of the single-class saving. It must also preserve the two mixed channels and the both-ultra channel from the exact cutoff difference. Fixed-form logarithmically averaged Elliott cancellation [1] is a useful benchmark, but it supplies neither an effective rate nor the required uniformity in the polynomially growing slopes, divisor families, rows, and masks.

7.4 Staged route after this paper

The resulting route is deliberately modular:

- TPC-31: prove determinant-entropy bounds for translated windows and individual normalized-determinant residue cells inside (89);
- TPC-32: seek signed dispersion across those cells, exploiting the factorable row source rather than only an arbitrary bounded mask;
- TPC-33: reassemble the mixed and both-ultra channels without separating away the cancellation encoded by the exact cutoff difference;
- TPC-34: return to the complete residual and endpoint packets only after the family estimate is uniform on the physical parameter box.

Failure at any one of these stages would still identify a sharp method boundary. None of these future interfaces is assumed in the theorems of the present paper.

Object	Status
Primitive target-gcd Euclidean geometry	Inherited from TPC-18 and restated exactly for opened rows.
Selected truncated packet through T	Closed under the TPC-28 physical conductor interfaces.
Large reflected content and content-rich sparse wedge	Closed in TPC-29.
Large full-target content $\mathcal{G} > Z$	Power-saving bound proved here by theorem 4.4 .
Large quotient or normalized cross-content	Power-saving bound proved here by corollary 5.3 .
Small full-target-content and fully coprime core	Open; next determinant-entropy and signed-dispersion gate.
Complete residual, positivity, prime-pair main term, or parity breach	Not proved and not implied.

8 Exact certificate, dependencies, and scope

8.1 Finite exact certificate

The standard-library script

`experiments/tpc30_certificate.py`

uses exception-based checks in ordinary and optimized Python. It verifies the finite algebra underlying the paper by independent direct and structured computations:

- (i) the primitive identity

$$(mj + h, nj + h) = (mj + h, m - n),$$

together with an explicit nonprimitive counterexample;

- (ii) exact- c orbit layers, direct occupancy in finite windows $\mathcal{I}$ of length $O_{\mathcal{D}}(J)$, and the fixed-row large-content inequality

$$\#\{j \in \mathcal{I} : (j, h) = 1, \mathcal{G}_{m,n}(j) > Z\} \leq \tau(|m - n|) \left(1 + \frac{J}{Z}\right);$$

- (iii) the four-factor identity

$$(rU, sV) = (r, s)(U, V)(r/(r, s), V/(U, V))(s/(r, s), U/(U, V));$$

- (iv) the explicit finite row-residue inequality

$$\mathcal{R}_c \leq K_c \|\gamma^{(1)}\|_2 \|\gamma^{(2)}\|_2,$$

where K_c is the largest row occupancy of a residue class;

- (v) the central rational high- β ledger, including the old zero saving and the relaxed margin 133/400;
- (vi) metadata regression assertions recording each full-residual, positivity, prime-pair, and parity claim deliberately excluded here.

The four-factor regression exhausts all 65,536 quadruples $1 \leq r, s, U, V \leq 16$ and separately checks 116 affine-target factorizations. Ordinary and optimized Python both complete 773,361 recorded top-level exact checks and regenerate byte-identical JSON. The subcounts are

- 441,578 primitive target-gcd identity checks;
- 1,924 exact-content occupancy checks;
- 1,135 fixed-row large-content checks;
- 197,313 exact four-factor checks;
- 131,301 row-residue energy checks;
- 90 rational high- β ledger checks;
- 20 scope and nonoverlap metadata regression assertions.

The reproducibility values are

- source SHA-256: f4813d6e89163098187fe1ce585934b354a384a09b7ca4332e3fad42d1a78610;
- JSON SHA-256: c466cd747c726d51074c0d8967a950ee0bcfb3e8a4658fcb8d3a81e52615c2e9;
- certificate digest: 8c52338642c27d2c4dafd6807cb8cc56d52888ae16b27cd9aeda880ccc10f64d.

The source hash normalizes source newlines to LF; the certificate digest hashes the canonical sorted payload before the digest field is inserted.

These computations certify finite identities, explicit finite inequalities, and rational arithmetic. They do not prove an asymptotic Möbius estimate. In particular, finite enumeration is not used as evidence that the high- β cell contains a positive amount of mass.

8.2 Dependency ledger

Input or statement	Status	Use
Opened-row ℓ^1/ℓ^2 provenance	Imported	TPC-25; pair mass and the auxiliary exact-layer energy.
Calibrated cutoff difference and pointwise divisor envelopes	Imported	TPC-28–TPC-29; defines the three raw channels.
Primitive target-gcd divisibility	Inherited geometry	TPC-18; sharpened here to the exact opened-row identity and orbit class.
Fixed-row large-content occupancy and weighted tail theorem	Proved here	Gives the $Q^2 + XQ/Z$ physical estimate.
Exact four-factor decomposition and cross-factor pruning	Elementary identity; pruning proved here	Reaches quotient-rich and cross-rich sectors.
TPC-29 reflected-content theorems	Imported for comparison	Establishes strict nonoverlap of the high- β sample.
Fixed-form logarithmic Elliott theorem	External benchmark	Not used in any positive estimate here.
Signed determinant-family dispersion	Open	Proposed next input for the present route.

8.3 Claims deliberately excluded

The following statements are not consequences of this paper:

- cancellation or closure on every small-target-gcd fiber;
- closure of the complete ultra-long cutoff difference or the full TPC-18 residual;
- an effective polynomially uniform Chowla–Elliott theorem;
- a positive lower bound, a Hardy–Littlewood asymptotic, or infinitely many twin primes;
- a general violation or weakening of the sieve parity barrier.

The positive conclusion is narrower: within the retained three-channel physical packet, every polynomially large full-target-content sector is power-small, including quotient-rich and cross-rich sectors that are invisible to reflected content alone. The remaining small-target-content residual is not solved; its fully coprime sector is identified as the sharp model.

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